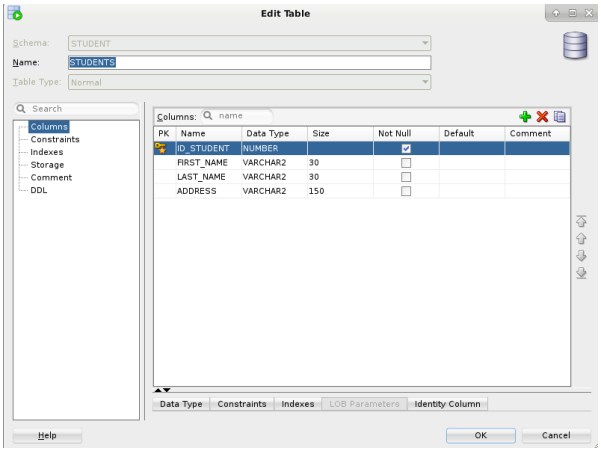
LAB 1: Basic operation on database

1. Creating tables

I created 3 tables: Students, Grades and Courses. I added table Students by choosing „new Table“ command and selecting everything I needed from the instructions. Editing table is shown on the picture on the right.



Other tables I added using SQL commands like this:

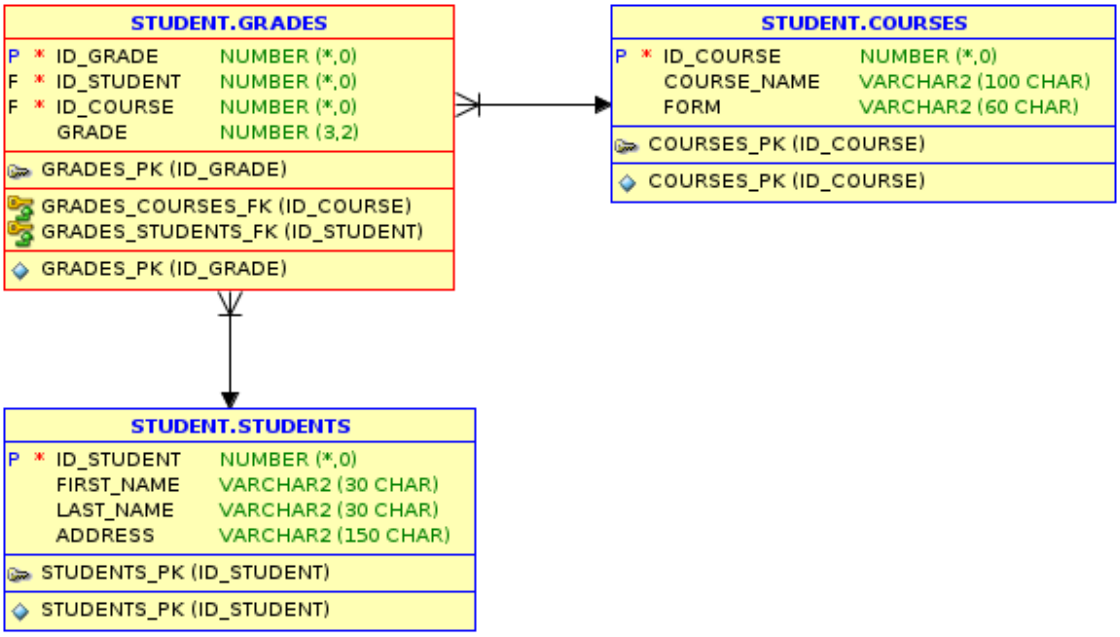
```
CREATE TABLE Courses (
 ID_course INTEGER NOT NULL,
 Course_Name VARCHAR2(100),
 Form VARCHAR2(60),
 CONSTRAINT Courses_PK PRIMARY KEY (ID_course)
);
```

```
CREATE TABLE Grades (
 ID_grade INTEGER NOT NULL,
 ID_student INTEGER NOT NULL,
 ID_course INTEGER NOT NULL,
 Grade NUMBER(3,2),
 CONSTRAINT Grades_PK PRIMARY KEY (ID_grade)
);
```

```
ALTER TABLE Grades
ADD CONSTRAINT Grades_Students_FK
FOREIGN KEY (ID_student)
REFERENCES Students(ID_student);
```

```
ALTER TABLE Grades
ADD CONSTRAINT Grades_Courses_FK
FOREIGN KEY (ID_course)
REFERENCES Courses(ID_course);
```

This is the ER diagram I now have:



2. Inserting new rows

a) Add five students to table student

```
INSERT INTO STUDENT.STUDENTS VALUES (1, 'Josip', 'Bosnic', 'Zagreb');
INSERT INTO STUDENT.STUDENTS VALUES (2, 'Anna', 'Zakharova', 'Poznan');
INSERT INTO STUDENT.STUDENTS VALUES (3, 'Blagica', 'Postolova', 'Stip');
INSERT INTO STUDENT.STUDENTS VALUES (4, 'Ivan', 'Tomic', 'Vodice');
INSERT INTO STUDENT.STUDENTS VALUES (5, 'Veronika', 'Budimir', 'Mostar');
```

c) Add four courses in two forms (8 records) to table Courses

```
INSERT INTO STUDENT.COURSES VALUES (1, 'Database Applications', 'LAB');
INSERT INTO STUDENT.COURSES VALUES (2, 'Database Applications', 'Lecture');

INSERT INTO STUDENT.COURSES VALUES (3, 'Physics', 'LAB');
INSERT INTO STUDENT.COURSES VALUES (4, 'Physics', 'Lecture');

INSERT INTO STUDENT.COURSES VALUES (5, 'Computer Science', 'LAB');
INSERT INTO STUDENT.COURSES VALUES (6, 'Computer Science', 'Lecture');

INSERT INTO STUDENT.COURSES VALUES (7, 'Innovation', 'LAB');
INSERT INTO STUDENT.COURSES VALUES (8, 'Innovation', 'Lecture');
```

These are the existing courses now:

ID_COURSE	COURSE_NAME	FORM
5	Computer Science	LAB
6	Computer Science	Lecture
1	Database Applications	LAB
2	Database Applications	Lecture
7	Innovation	LAB
8	Innovation	Lecture
3	Physics	LAB
4	Physics	Lecture

b) In table Grades - for each student add one grade for each course and each form

For student 1:

```
INSERT INTO STUDENT.GRADES (ID_GRADE, ID_STUDENT, ID_COURSE, GRADE)
VALUES (1, 1, 1, 4);
INSERT INTO STUDENT.GRADES VALUES (2, 1, 2, 5);
INSERT INTO STUDENT.GRADES VALUES (3, 1, 3, 3);
INSERT INTO STUDENT.GRADES VALUES (4, 1, 4, 4);
INSERT INTO STUDENT.GRADES VALUES (5, 1, 5, 5);
INSERT INTO STUDENT.GRADES VALUES (6, 1, 6, 3);
INSERT INTO STUDENT.GRADES VALUES (7, 1, 7, 4);
INSERT INTO STUDENT.GRADES VALUES (8, 1, 8, 5);
```

For student 2:

```
INSERT INTO STUDENT.GRADES VALUES (9, 2, 1, 3);
INSERT INTO STUDENT.GRADES VALUES (10, 2, 2, 4);
INSERT INTO STUDENT.GRADES VALUES (11, 2, 3, 5);
INSERT INTO STUDENT.GRADES VALUES (12, 2, 4, 3);
INSERT INTO STUDENT.GRADES VALUES (13, 2, 5, 4);
INSERT INTO STUDENT.GRADES VALUES (14, 2, 6, 5);
INSERT INTO STUDENT.GRADES VALUES (15, 2, 7, 3);
INSERT INTO STUDENT.GRADES VALUES (16, 2, 8, 4);
```

For student 3:

```
INSERT INTO STUDENT.GRADES VALUES (17, 3, 1, 5);
INSERT INTO STUDENT.GRADES VALUES (18, 3, 2, 4);
INSERT INTO STUDENT.GRADES VALUES (19, 3, 3, 3);
INSERT INTO STUDENT.GRADES VALUES (20, 3, 4, 5);
INSERT INTO STUDENT.GRADES VALUES (21, 3, 5, 4);
INSERT INTO STUDENT.GRADES VALUES (22, 3, 6, 3);
INSERT INTO STUDENT.GRADES VALUES (23, 3, 7, 5);
INSERT INTO STUDENT.GRADES VALUES (24, 3, 8, 4);
```

For student 4:

```
INSERT INTO STUDENT.GRADES VALUES (25, 4, 1, 4);
INSERT INTO STUDENT.GRADES VALUES (26, 4, 2, 3);
INSERT INTO STUDENT.GRADES VALUES (27, 4, 3, 5);
INSERT INTO STUDENT.GRADES VALUES (28, 4, 4, 4);
INSERT INTO STUDENT.GRADES VALUES (29, 4, 5, 3);
INSERT INTO STUDENT.GRADES VALUES (30, 4, 6, 5);
INSERT INTO STUDENT.GRADES VALUES (31, 4, 7, 4);
INSERT INTO STUDENT.GRADES VALUES (32, 4, 8, 3);
```

For student 5:

```
INSERT INTO STUDENT.GRADES VALUES (33, 5, 1, 3);
INSERT INTO STUDENT.GRADES VALUES (34, 5, 2, 5);
INSERT INTO STUDENT.GRADES VALUES (35, 5, 3, 4);
INSERT INTO STUDENT.GRADES VALUES (36, 5, 4, 3);
INSERT INTO STUDENT.GRADES VALUES (37, 5, 5, 5);
INSERT INTO STUDENT.GRADES VALUES (38, 5, 6, 4);
INSERT INTO STUDENT.GRADES VALUES (39, 5, 7, 3);
INSERT INTO STUDENT.GRADES VALUES (40, 5, 8, 5);
```

3. Data filtering

a) Show all students in lexicographical order (ascending or descending)

b) Show all courses (in lexicographical order) divided into groups based on course forms

c) Show all grades in form : First_name Last_Name, grade, Course_name, form

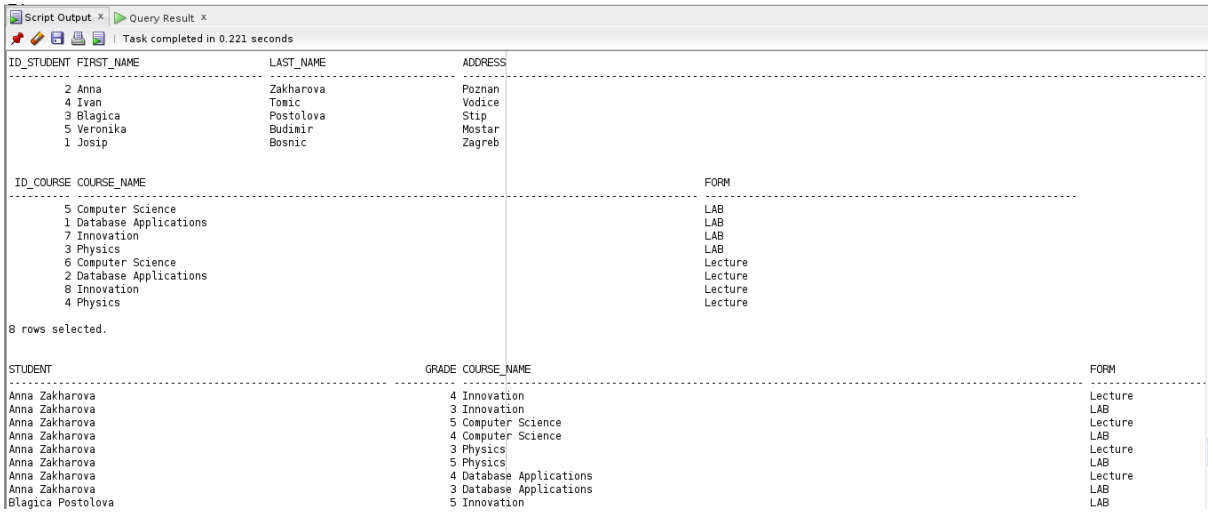
For a b and c I used these SQL commands:

```
SELECT *
FROM STUDENT.STUDENTS
ORDER BY LAST_NAME DESC;
```

```
SELECT *
FROM STUDENT.COURSES
ORDER BY FORM ASC, COURSE_NAME ASC;
```

```
SELECT
   s.FIRST_NAME || ' ' || s.LAST_NAME AS STUDENT,
   g.GRADE,
   c.COURSE_NAME,
   c.FORM
FROM STUDENT.GRADES g
JOIN STUDENT.STUDENTS s
   ON g.ID_STUDENT = s.ID_STUDENT
JOIN STUDENT.COURSES c
   ON g.ID_COURSE = c.ID_COURSE
ORDER BY STUDENT;
```

Part of the output is shown on the picture below:



Script Output | Query Result

Task completed in 0.221 seconds

ID_STUDENT	FIRST_NAME	LAST_NAME	ADDRESS
2	Anna	Zakharova	Poznan
4	Ivan	Tomic	Vodice
3	Blagica	Postolova	Stip
5	Veronika	Budimir	Mostar
1	Josip	Bosnic	Zagreb

ID_COURSE	COURSE_NAME	FORM
5	Computer Science	LAB
1	Database Applications	LAB
7	Innovation	LAB
3	Physics	LAB
6	Computer Science	Lecture
2	Database Applications	Lecture
8	Innovation	Lecture
4	Physics	Lecture

8 rows selected.

STUDENT	GRADE	COURSE_NAME	FORM
Anna Zakharova	4	Innovation	Lecture
Anna Zakharova	3	Innovation	LAB
Anna Zakharova	5	Computer Science	Lecture
Anna Zakharova	4	Computer Science	LAB
Anna Zakharova	3	Physics	Lecture
Anna Zakharova	5	Physics	LAB
Anna Zakharova	4	Database Applications	Lecture
Anna Zakharova	3	Database Applications	LAB
Blagica Postolova	5	Innovation	LAB

4. Data modifications

a) Delete all grades belonging to course which has the smallest average

First I calculated averages for all courses:

```
SELECT
   c.COURSE_NAME,
   AVG(g.GRADE) AS AVERAGE_GRADE
FROM STUDENT.GRADES g
JOIN STUDENT.COURSES c
   ON g.ID_COURSE = c.ID_COURSE
GROUP BY c.COURSE_NAME
ORDER BY AVERAGE_GRADE ASC;
```

COURSE_NAME	AVERAGE_GRADE
Physics	3.9
Innovation	4
Computer Science	4.1
Database Applications	4.2

From the output we can see that Physics has the lowest average grade so I am deleting it using these SQL commands:

```
DELETE FROM STUDENT.GRADES
WHERE ID_COURSE IN (
  SELECT ID_COURSE
  FROM STUDENT.COURSES
  WHERE COURSE_NAME = 'Physics'
);
```

b) Reduce by 0.5 all grades belonging to course which has the greatest average

From the output image we can see that the course Database Applications has the greatest average.

```
UPDATE STUDENT.GRADES
SET GRADE = GRADE - 0.5
WHERE ID_COURSE IN (
  SELECT ID_COURSE
  FROM STUDENT.COURSES
  WHERE COURSE_NAME = 'Database Applications'
);
```

When we calculate the averages again we can see that there are now only 3 courses left and that the average for Database Applications is 0.5 lower than before.

COURSE_NAME	AVERAGE_GRADE
Computer Science	4.1
Innovation	4
Database Applications	3.7